SMART FACIAL SCREENING FOR DS DETECTION

Automated AI-assisted screening - not a diagnosis

Patient Information	
Name	ma
Date	2026-03-04 13:36:53
Child Age	13
Mother Age	36
Father Age	43
Parents Relation	Distant Relative
Living Area	Urban Area
Previous Pregnancies	0
Screening Summary	
Probability	85%
Confidence	85%
Risk Level	High
High Confidence	No

Facial Features (AI Analysis)	
Facial Symmetry	0.86
Eye Spacing	0.61
Nasal Bridge	0.70
Ear Position	0.56

AI Explanation

{ "title": "**Down Syndrome Risk: Eyes, Nose, and Mouth**", "explanation": "The patient's Grad-CAM heatmap shows significant activation in the eyes, nose, and mouth regions, indicating potential Down syndrome traits. The eyes have a high activation score of 0.628, suggesting a higher risk. The nose has an activation score of 0.606, and the mouth has a score of 0.433. These regions are crucial for facial recognition and speech, which could impact the child's development. The current risk level is high, with a probability of 85%. The affected regions are likely to influence future health issues such as speech delay and feeding difficulties. The child should monitor these areas closely.", "affected_regions": ["eyes", "nose", "mouth"], "food_recommendations": ["Protein-rich foods", "Vitamin C-rich fruits", "Iron-rich foods"], "exercise_recommendations": ["Head bobbing exercises", "Eye tracking exercises", "Mouth opening exercises"],

"video_titles": ["Down Syndrome Awareness", "Facial Recognition Exercises", "Speech Therapy Tips"], " Risk overview: High likelihood (85%) based on Grad-CAM facial pattern analysis. Affected facial regions: - eyes: score 0.628, severity high. Peri-orbital spacing/alignment patterns were highlighted in this region. - nose: score 0.606, severity high. Mid-face and nasal-bridge structure patterns were highlighted in this region. - mouth: score 0.433, severity moderate. Lower mid-face/oral contour patterns were highlighted in this region. Risk factors (Grad-CAM): eyes: 0.628 (high, 27.4%); nose: 0.606 (high, 26.4%); mouth: 0.433 (moderate, ...

Recommendations

Food

- "Protein-rich foods"
- "Vitamin C-rich fruits"
- "Iron-rich foods"
- Omega-3 rich foods
- leafy greens
- fruits

Exerc

- "Head bobbing exercises"
- "Eye tracking exercises"
- "Mouth opening exercises"
- Eye focus drills
- gentle neck stretches
- breathing
- supervised physiotherapy

Exer

- Gentle Child Mobility Routine
- Beginner Facial Muscle Exercise Session

Detailed Result Fields

Risk Factor Explanation: eyes: score 0.628, severity high, contribution 27.4%; nose: score 0.606, severity high, contribution 26.4%; mouth: score 0.433, severity moderate, contribution 18.9%; forehead: score 0.337, severity low, contribution 14.7%; facial_symmetry: value 0.86, severity high; eye_spacing: value 0.61, severity high; nasal_bridge: value 0.7, severity high; ear_position: value 0.56, severity moderate

Affected Region Explanation: eyes: score 0.628, severity high. Peri-orbital spacing/alignment patterns were highlighted in this region. nose: score 0.606, severity high. Mid-face and nasal-bridge structure patterns were highlighted in this region. mouth: score 0.433, severity moderate. Lower mid-face/oral contour patterns were highlighted in this region.

Future Health Explanation: Speech and language delay [high]: Consider early speech and language support. Low muscle tone [high]: Track posture, motor milestones, and physiotherapy needs. Feeding or swallowing difficulty [high]: Monitor feeding tolerance and growth trend. Slower developmental milestones [high]: Use periodic developmental screening follow-up. Oral-motor articulation concern [high]: Watch speech clarity and oral-motor control. Breathing/sleep quality observation [high]: Track persistent nasal obstruction or sleep issues. Vision alignment observation [high]: Schedule eye check if alignment concerns appear. Maternal age risk context [high]: Ensure continued pediatric/genetic follow-up due to advanced maternal age context.

Food Explanation: "Protein-rich foods",; "Vitamin C-rich fruits",; "Iron-rich foods"; Omega-3 rich foods; leafy greens; fruits

Exercise Explanation: "Head bobbing exercises",; "Eye tracking exercises",; "Mouth opening exercises"; Eye focus drills; gentle neck stretches; breathing

Risk Factors:
- facial_symmetry [high]
- nasal_bridge [high]
- eyes [high]
- eye_spacing [high]

Affected Region Details:
- eyes [high]: Peri-orbital spacing/alignment patterns were highlighted in this region.
- nose [high]: Mid-face and nasal-bridge structure patterns were highlighted in this region.
- mouth [moderate]: Lower mid-face/oral contour patterns were highlighted in this region.

Future Health Issues:
- Speech and language delay [high]: Consider early speech and language support.
- Low muscle tone [high]: Track posture, motor milestones, and physiotherapy needs.
- Feeding or swallowing difficulty [high]: Monitor feeding tolerance and growth trend.
- Slower developmental milestones [high]: Use periodic developmental screening follow-up.
- Oral-motor articulation concern [high]: Watch speech clarity and oral-motor control.
- Breathing/sleep quality observation [high]: Track persistent nasal obstruction or sleep issues.
- Vision alignment observation [high]: Schedule eye check if alignment concerns appear.
- Maternal age risk context [high]: Ensure continued pediatric/genetic follow-up due to advanced maternal age context.

This report is for screening purposes only and is not a diagnosis.